Project Design Phase-I
Solution Architecture

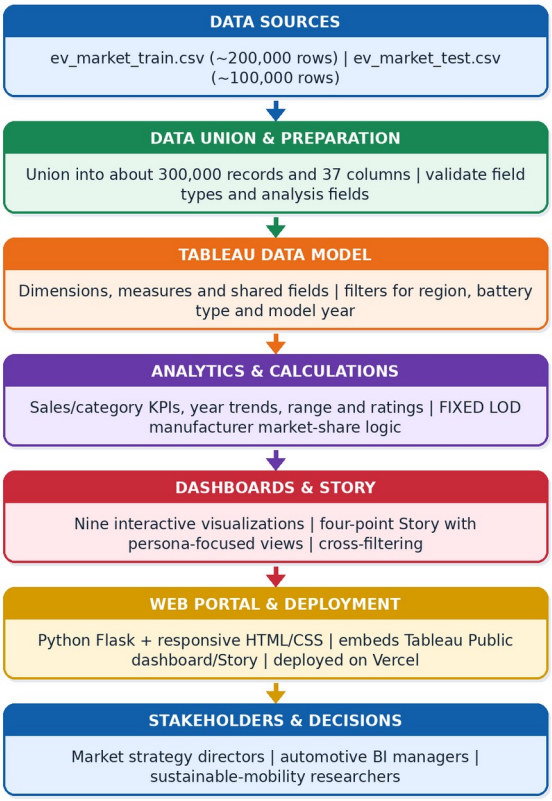
Date	28 October 2026
Team ID	Individual Project - Ujjawal Rathod
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

EV Market Analytics Architecture:



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data sources and integration	Combines the training and test CSV sources into one analysis dataset of about 300,000 records and 37 columns.	CSV files; Tableau data connection / union
Data preparation	Organizes dimensions and measures for consistent comparisons and checks the fields selected for analysis.	Tableau data pane; calculated fields
Analytics and calculations	Supports sales, category, model-year, manufacturer, battery, range, ratings and geography analysis. FIXED LOD logic is used for manufacturer market share.	Tableau calculated fields; FIXED LOD
Interactive dashboard controls	Enables exploration by market region, battery type and model year; dashboard actions connect related visualizations.	Tableau filters and dashboard actions
Dashboards and Story	Presents nine visualizations and a four-point Story with persona-focused views for market strategy, automotive BI and sustainability research.	Tableau Public/Desktop
Web portal and deployment	Embeds the Tableau Public dashboard and Story in a responsive web application so reviewers can access the analytics in a browser.	Python Flask; HTML/CSS; Vercel
Version control and documentation	Maintains source code and project documentation for review and final submission.	Git; GitHub repository